BFL: A User's Decision Guide

How `run_BFL()` picks the model from *how you build your inputs*

1 Setup and notation

A **target** site **T** (the one we want cause fractions for) and three fully labeled **source** sites **A**, **B**, **C**. T is only *partly* labeled (say 20% labeled, 80% unlabeled).

Symbol	Meaning
T; A, B, C	target site; three fully labeled source sites
T_X, T_Y	target feature matrix / cause labels
$A_X, \ldots, C_Y$	source feature matrices / cause labels
labeled rows	the small labeled part of T
unlabeled rows	the rest of T — **we always score here**
T self-site	a model trained on T's *labeled* rows

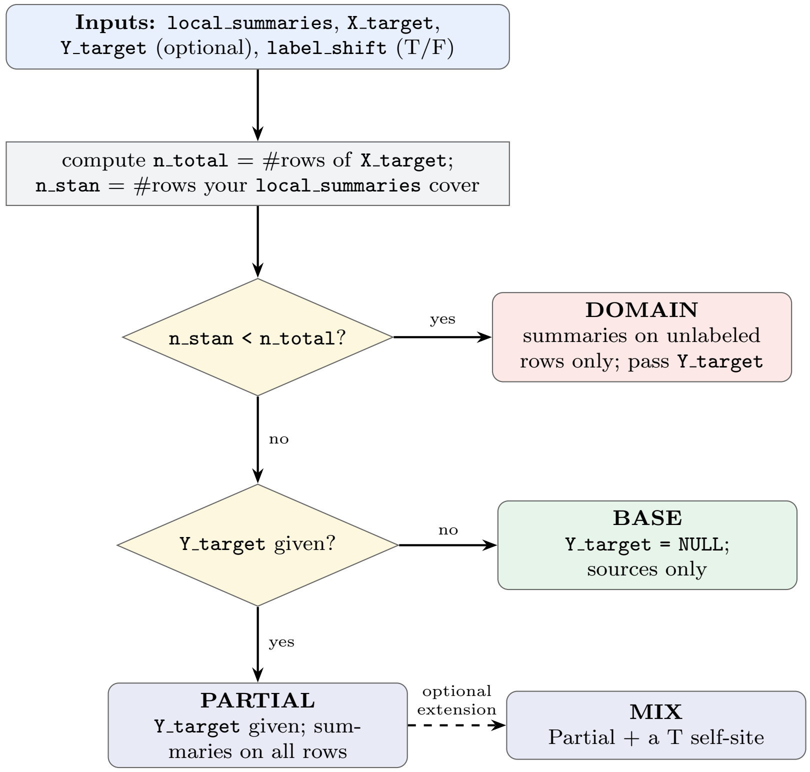
- `run_BFL()` has **no** "model type" argument — it *infers* the variant from the shape of your inputs.
- The hinge: do your `local_summaries` cover *all* target rows or only the *unlabeled* ones, and did you pass `Y_target`?
- Every variant is **scored on the unlabeled rows**; the variants differ in what extra information they feed the model.

2 Stage 0 — one-time setup (before any variant logic)

1. `X_target` $\rightarrow$ numeric matrix.
2. `n_total` $\leftarrow$ number of rows of `X_target`.
3. `ref_hashes` $\leftarrow$ one fingerprint per target row.
4. `n_stan` $\leftarrow$ number of rows your `local_summaries` cover.

The comparison `n_stan` vs. `n_total` drives the whole decision.

3 The decision tree



`label_shift` **(T/F) — set on every run; chooses the CSMF target.**

- `FALSE`: score against the *whole* target. When labels enter Stan (Partial/Mix) the `balanced` model is used; Domain folds the known labeled counts n_{Lc} back in.
- `TRUE`: score against the *unlabeled rows only*; Partial/Mix use the `unbalanced` model.

Base is unaffected — no labels reach Stan, so it is always `no_partial`, scored on the whole target.

Domain check: summaries must cover *exactly* the unlabeled rows (`#NA(Y_target) == n_stan`), else `run_BFL` errors.

Partial vs. Mix is not a branch. `run_BFL` can't tell them apart — **Mix is just Partial with one extra model** (a T self-site) added to `local_summaries`.

4 Building `local_summaries`

Each entry is one model's per-record scores (`posterior_phi`, `cause_ids`, `row_hash`). "T self-site" = an LCVA trained on T's *labeled* rows, then used to produce phi.

Variant	**Models in** `local_summaries`	**Summaries on target rows**	**Predict / score on**	`Y_target`
Base	sources A, B, C	all rows	unlabeled rows	`NULL`
Domain	sources A, B, C + T self-site	unlabeled only	unlabeled rows	labels (`NA` on unlab.)
Partial	sources A, B, C	all rows	unlabeled rows	labels (`NA` on unlab.)
Mix	sources A, B, C + T self-site	all rows	unlabeled rows	labels (`NA` on unlab.)

Base — sources only, no labels

```
ls <- list(A = sumA, B = sumB, C = sumC) # phi on ALL target rows
run_BFL(ls, X_target = T_X, Y_target = NULL, label_shift = FALSE)
```

Domain — sources + T self-site, on the unlabeled rows

```
selfFit <- LCVA.train(X = T_X_lab, Y = T_Y_lab) # T's labeled rows
sumT <- summarise(predict_phi(selfFit, T_X_unlab))# phi on unlabeled rows
ls <- list(A = sumA_unlab, B = sumB_unlab, C = sumC_unlab, T = sumT)
Y <- T_Y; Y[unlabeled] <- NA
run_BFL(ls, X_target = T_X, Y_target = Y, label_shift = FALSE)
```

Partial — sources only, labels fed to the model

```
ls <- list(A = sumA, B = sumB, C = sumC) # phi on ALL rows
Y <- T_Y; Y[unlabeled] <- NA
run_BFL(ls, X_target = T_X, Y_target = Y, label_shift = FALSE)
```

Mix — Partial *plus* a T self-site (the only change)

```
selfFit <- LCVA.train(X = T_X_lab, Y = T_Y_lab)
sumT <- summarise(predict_phi(selfFit, T_X)) # phi on ALL rows
ls <- list(A = sumA, B = sumB, C = sumC, T = sumT) # <-- extra model
Y <- T_Y; Y[unlabeled] <- NA
run_BFL(ls, X_target = T_X, Y_target = Y, label_shift = TRUE)
```

5 Summary: the arguments you decide

Argument	What it is	Your options
`local_summaries`	list of per-model phi (one per site)	sources A,B,C; add a T self-site for Domain/Mix
`X_target`	full target feature matrix	always all N rows
`Y_target`	target labels	`NULL` (Base); or labels with `NA` on unlabeled rows
`label_shift`	CSMF target	`FALSE` = whole target; `TRUE` = unlabeled subset

What you build	**run_BFL infers**
`Y_target = NULL`, summaries on all rows	`Base (no_partial)`
`Y_target` given, summaries on unlabeled rows + T self-site	`Domain (no_partial +` n_{Lc}`)`
`Y_target` given, summaries on all rows	`Partial (balanced / unbalanced)`
... plus a T self-site	`Mix (balanced / unbalanced)`